

Fourier Transform and the Poisson Kernel

Upper Half-Plane

Goal

- ▶ Define the Fourier transform via $L \rightarrow \infty$
- ▶ Solve Laplace's equation in the upper half-plane
- ▶ Recover the Poisson kernel

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Domain:

$$y > 0, \quad u(x, 0) = f(x)$$

Fourier Series on $[-L, L]$

Let f be defined on $[-L, L]$.

$$f(x) = \sum_{n=-\infty}^{\infty} c_n e^{in\pi x/L}$$

$$c_n = \frac{1}{2L} \int_{-L}^L f(x) e^{-in\pi x/L} dx$$

Frequency Interpretation

Define:

$$k_n = \frac{n\pi}{L}$$

Then:

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Spacing:

$$\Delta k = \frac{\pi}{L}$$

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As $L \rightarrow \infty$:

- ▶ $\Delta k \rightarrow 0$
- ▶ Sum becomes an integral

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$$\sum c_n e^{ik_n x} \longrightarrow \int_{-\infty}^{\infty} \hat{f}(k) e^{ikx} dk$$

Fourier Transform Pair

Define:

$$\hat{f}(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx$$

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Continuous analogue of Fourier series.

Problem Setup

Solve:

$$u_{xx} + u_{yy} = 0, \quad y > 0$$

$$u(x, 0) = f(x)$$

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Take Fourier transform in x .

Transforming the PDE

Let:

$$\hat{u}(k, y) = \int u(x, y) e^{-ikx} dx$$

Then:

$$\widehat{u_{xx}} = -k^2 \hat{u}$$

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So PDE becomes:

$$-k^2 \hat{u} + \frac{\partial^2 \hat{u}}{\partial y^2} = 0$$

Solve the ODE

$$\hat{u}_{yy} = k^2 \hat{u}$$

General solution:

$$\hat{u}(k, y) = A(k)e^{|k|y} + B(k)e^{-|k|y}$$

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Boundedness as $y \rightarrow \infty$ gives:

$$\hat{u}(k, y) = \hat{f}(k)e^{-|k|y}$$

Inverse Transform

$$u(x, y) = \int_{-\infty}^{\infty} \hat{f}(k) e^{-|k|y} e^{ikx} dk$$

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Interpret as convolution:

$$u(x, y) = (P_y * f)(x)$$

Kernel Definition

Define:

$$P_y(x) = \int_{-\infty}^{\infty} e^{-|k|y} e^{ikx} dk$$

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We compute this explicitly.

Split the Integral

$$P_y(x) = \int_0^\infty e^{-ky} e^{ikx} dk + \int_0^\infty e^{-ky} e^{-ikx} dk$$

Split the Integral

$$\begin{aligned}P_y(x) &= \int_0^\infty e^{-ky} e^{ikx} dk + \int_0^\infty e^{-ky} e^{-ikx} dk \\&= \int_0^\infty e^{-k(y-ix)} dk + \int_0^\infty e^{-k(y+ix)} dk\end{aligned}$$

Evaluate Integrals

$$\int_0^{\infty} e^{-k(y \mp ix)} dk = \frac{1}{y \mp ix}$$

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Thus:

$$P_y(x) = \frac{1}{y - ix} + \frac{1}{y + ix}$$

Simplify

$$P_y(x) = \frac{2y}{x^2 + y^2}$$

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Including normalization:

$$P_y(x) = \frac{1}{\pi} \frac{y}{x^2 + y^2}$$

Final Poisson Integral Formula

$$u(x, y) = \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{y}{(x - \xi)^2 + y^2} f(\xi) d\xi$$

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This is the Poisson kernel for the upper half-plane.

Connection to Theorem 10.3.1

The Poisson kernel:

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Exactly matches Theorem 10.3.1.

Key Takeaways

- ▶ Fourier transform arises as $L \rightarrow \infty$
- ▶ PDE reduces to algebra in frequency space
- ▶ Exponential decay $e^{-|k|y}$ encodes harmonicity
- ▶ Inverse transform gives Poisson kernel